

Research article

Invisible touch sensors-based smart and disposable door locking system for security applications

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ABSTRACT

Nowadays, security is one of the living essentials, and there is a dire need for reliable, secure, and smarter locking systems. The stand-alone smart security systems are of great interest as they do not involve keys, cards, or unsecured communication in order to prevent carrying, loss, duplication, and hacking. Here, we report an invisible touch sensors-based smart door locking system (DLS). The passive transducer-based touch sensors are fabricated through a facile do-it-yourself (DIY) based fabrication process by pasting the hybrid geometry copper electrodes on cellulose paper. The employment of biodegradable, and non-toxic materials like paper and copper tape makes this configuration a good candidate for green electronics. For additional security, the keypad in the DLS is made invisible by covering it with paper and spray paint. One can only open the door by knowing the password as well as the location of each key on the sensor keypad. The system can efficiently recognize the exact pattern of passwords without any false values. Invisible touch sensors-based locking systems can easily contribute to the security applications in homes, banks, automobiles, apartments, lockers, and cabinets.

1. Introduction

Security is one of the most crucial parts of our houses, banks, corporations, and automobiles. Usually, people ensure their safety through physical key-based locks. However, issues like carrying, duplication, and loss are associated with physical keys [1]. Modern-day door lock systems are either button-pressed smart lock systems or radio frequency identification (RFID) card-based locking systems [2]. Smart homes use a button or touch keypad-based interface for digital door locks. The functionality of such keypad-based digital locks can be increased through unique programmable passwords and this approach is highly useful for hotels and rental apartments. However, the fingerprint on touchscreens may provide clues about the password.

Technological advancements are rapidly changing the world security prospect because of the immense use of electronic devices like cameras, mobile phones, Bluetooth and Wi-Fi-controlled locks, human-machine interfaces (HMIs), and IoT devices [3–7]. However,

Abbreviations: DLS, Door Locking System; HMI, Human Machine Interface; DIY, Do-It-Yourself; CDC, Charge-Discharge Cycles.

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Fig. 1. Schematic illustration of smart touch sensors working for DLS.

this increase in the use of electronic devices is causing an exponential increase in the amount of electronic waste (e-waste). This e-waste is causing severe environmental problems [4]. It's estimated that the Waste Electrical and Electronic Equipment (WEEE) and e-waste generated in 2021 was 57.5 million metric tons (Mt) [8]. About 42% of WEEE and e-waste comprises waste printed circuit boards (WPCB). According to the 2019 formal documentation, the recycling of WEEE was ~17%, and a significant part of WEEE ~83% was not appropriately managed [9]. Paper and other disposable substrates are useful alternatives to develop WEEE-free electronic devices and sensors [10]. Along with e-waste issues, digital wireless locks have an implicit issue of unsecured communication as well.

The attractions of disposable, flexible, and throw-away devices are increasing due to their high potential in human-machine interfaces, smart packaging, health care, etc [11–15]. Developing disposable and flexible devices and sensors involves flexible substrates and conductive inks. Commonly used substrates such as polyimide and polyethylene terephthalate are essential for flexible electronics, but these substrates are also not biodegradable and can cause e-waste [16,17]. Alternate substrate materials, especially cellulose-based printing paper, is getting attention due to their biodegradability, flexibility and easy availability [18]. Paper substrate-based studies have been reported for a wide range of applications including gas sensors [19], HMIs [4], glucose sensors [20], humidity sensors [21], supercapacitors [22], transistors [23], touch sensors [11,24], and solar cells [25]. Along with the substrate, the development process of conductive inks involves solvents and chemical-based costly and hazardous procedures. The fabrication of commercially available Bluetooth, Wi-Fi-controlled lock, and IoT devices also involve complex, cleanroom-based, and energy-consuming processes resulting in greenhouse gas emissions [26–30]. In this regard, an environment-friendly fabrication process is required that involves the easily available and facile materials for developing disposable DLS.

Owing to the e-waste and secrecy issues, we report a facile and green electronics-based invisible touch sensors system. The sensors were fabricated using low-cost, e-waste-free, and non-toxic materials like paper and copper tape. Five sensors consisting of numeric and alphabetic keys based on hybrid geometry were fabricated in this work. The numeric keys were used to enter the password, and the alphabetic key R was used to reset the password. The sensors are more reliable compared to the plastic keys-based keypad. To make the sensor's keypad invisible, it was covered with paper or paints. The sensors work proficiently in invisible configuration as well—keypad's invisibility is an added advantage for security applications. One cannot open the door unless he knows the password as well as the location of each key on the sensor keypad. This feature provides additional security in case of a password leak. A schematic illustration of practicability is shown in Fig. 1 by showing sensors on a door lock.

2. Fabrication and experimental setup

Sensors were fabricated using low-cost and easily available materials i.e. paper and copper tape using solvent-free, garage, and DIY fabrication [24]. 80-gram A4 printer paper was utilized to print the MS word designed sensors. Copper tape was cut and pasted according to the hybrid geometry based numerics 1, 2, 3, 4 and alphabet R. No insulated copper tape with non-toxic stickiness with 0.5 mm was utilized to fabricate the sensors. The sensors were connected with the ESP microcontroller module using the copper-based jumper wires. The invisibility was achieved by covering the sensors with paper initially to check the working of sensors and then painted through 4–5 layers of market-available white spray paint. The four sensors allow the users to choose any password from 24 possible combinations. In addition, the system can generate an alarm if someone enters the wrong password twice or tries to harm the

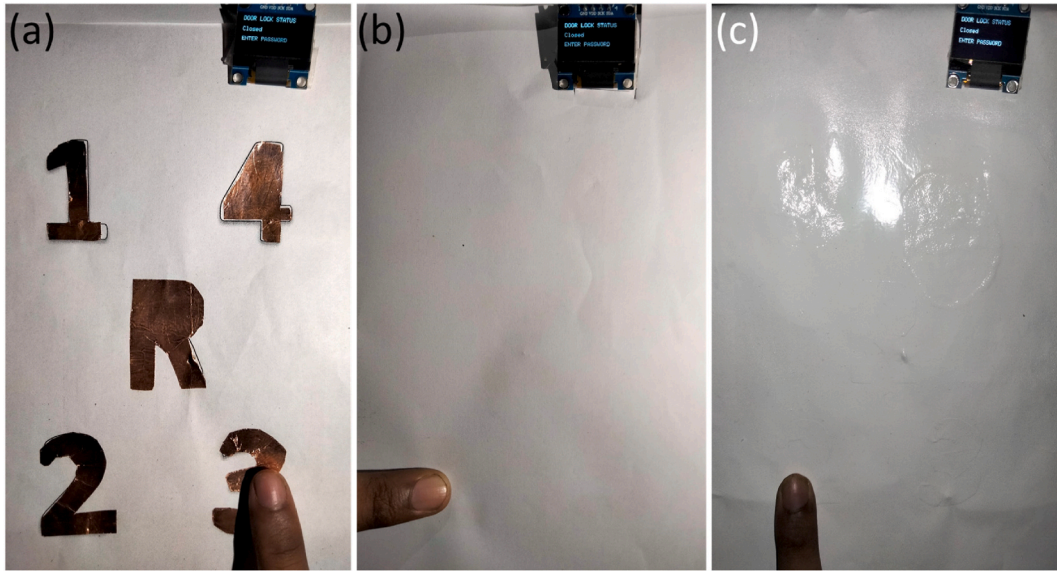


Fig. 2. DIY fabricated sensors and experimental setup. (a) Hybrid geometry-based sensors fabricated using copper tape. (b). Sensors under test while invisible using paper on sensors. (c) Sensors testing while invisible using white spray paint coatings.

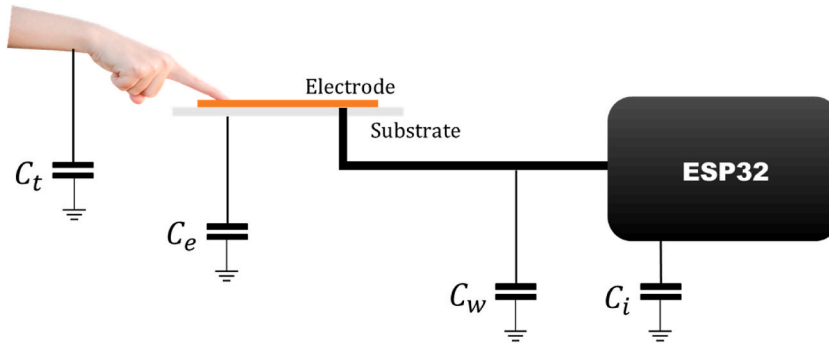


Fig. 3. Circuit diagram of working of sensors with ESP32.

sensors. Fig. 2 (a) shows the fabricated sensors system consisting of numeric and alphabetic keys and OLED to show the door status and guide the user about what to do next in case of an open door or change password following the developed algorithm. The pictures of sensors working invisibly under the paper and white spray paint are shown in Fig. 2 (b) and 2 (c) respectively.

Capacitive touch sensor normally responds to finger touch by measuring the capacitance of two electrode. Human finger interaction with the sensor surface acts as a parallel plate capacitor and the capacitance of touch is known as touch capacitance and shown in Fig. 3 as C_t . Total capacitance of the circuit interfacing the touch sensor with ESP module is C_{total} and consists of internal capacitance of the module C_i , capacitance of the wire C_w , capacitance of the electrode C_e and the touch capacitance C_t [24].

$$C_{total} = C_i + C_w + C_e + C_t \quad (1)$$

The touch capacitance of sensor changed with the human touch to the sensor. C_i , C_w , and C_e are content values capacitances and known as peracetic capacitance C_p . When touch exerts at electrode, C_t introduced as parallel with C_p that will change the net capacitance. As a result, RC constant has new value that alters the charging discharging time for the capacitor. The increase in the capacitance increases the RC time constant and it decrease the number of charging and discarding cycles ESP module from usual circuit without touch. Th touch increases the overall capacitance of the circuits and it decrease the number of measurements cycles of FSM of the ESP module.

3. Results and discussion

Passive transducer-based touch sensors are reported here that measure the finger's touch response by capacitance measurement between two parallel plates. The touch on the sensor surface formed a parallel plate between the finger and the sensor's surface. The

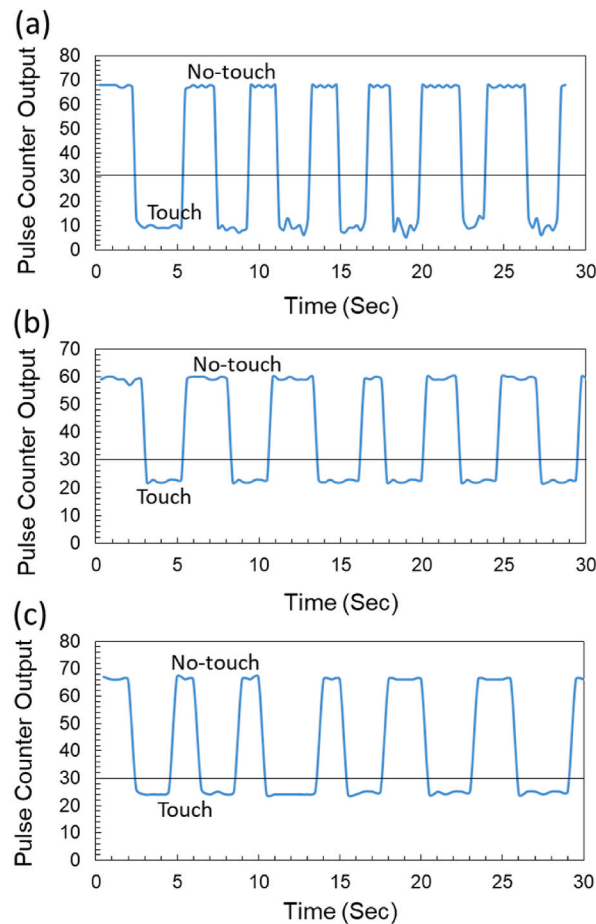


Fig. 4. CDC response of single touch sensor with time. (a) Bare sensor response in touch and no-touch conditions with time. (b) Paper and (c) spray paint shielded sensor CDC response with time.

finger touch changes the capacitance because of the body's mass acting as an earth ground through the figure conductivity. The ESP module that is connected with the sensor measures the charge-discharge cycles (CDC) of touch sensors. FSM of the ESP module scans the charging and discharging of capacitances interfaced with it. The pulse counter of the module counts the response obtained from the charging and discarding cycles for specific times and shown it in between 0 and 80. At maximum it counts as maximum number of cycles as 80 with minimum capacitance interfaced with the module. As the capacitance increased in connection with the module the number of cycles reduced because of large RC time constant. The sensors response for finger touch on the bare sensor is shown in Fig. 4 (a). Touch sensors were tested multiple times for touch and no-touch states to see the ESP FSM counter values to decide the threshold. Fig. 4 (b) and (c) show, respectively, the counter values of CDC of paper invisible and paint invisible touch sensors in touch and no-touch states. The fabricated sensor showed consistent results with zero false positives for the sensors in visible and invisible situations with the same threshold value.

Four numeric sensors 1, 2, 3, 4 were fabricated to enter the password. In addition, one alphabetic sensor R was fabricated to utilize for password reset. All the five touch sensors were fabricated and tested with ESP to check their working. Fig. 5 shows the CDC response of all the sensors in bare and invisible configurations. The threshold was set at 30 and all the sensors worked properly. Fig. 5 (a) shows the response of all the touch sensors in the bare scenario. Fig. 5 (b) and (c) show the sensor's response of all five sensors after paper and spray paint cover, respectively. It can be seen that the performance of sensor is unaffected by the paper and spray coating.

An algorithm was also developed to operate the DLS. The flow chart of the developed algorithm is shown in Fig. 6. The system is always ready to collect the entered password. When the user enters the password through touch sensors, the system opens the door if the password matches the pre-set password. Supplementary Video S1 demonstrates the working of the sensors system for door unlocking using the correct pattern as 1,2,3,4. If the entered password does not match with the pre-set password, the door will remain closed. The password reset option functioned with a touch entry of R. After the reset request, the system would ask for last password and would not allow to reset the password until the previous password has been entered correctly. After the correct entry of the last password, the system would allow entering the new password. Supplementary Video S2 demonstrate the working of the system to reset the password using the R key from old password 1,2,3,4 to a new password 1,2,1,2. The working of the sensors is also demonstrated in Supplementary Video S3 by touching the surface other than the sensors and system don't detect it as any input and then correct

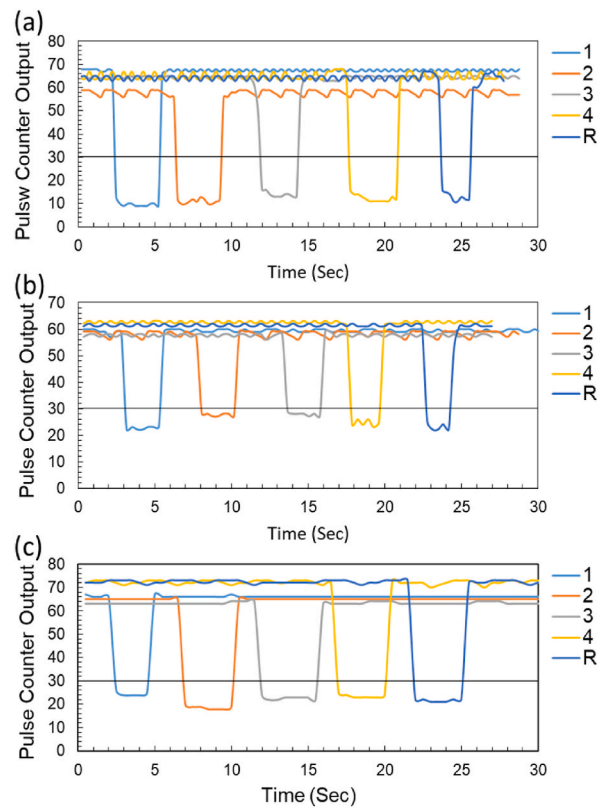


Fig. 5. CDC response of all the touch sensors 1, 2, 3, 4, and R with time. (a) Bare sensors response to touch and no-touch conditions with time. (b) Paper and (c) spray paint shielded sensors CDC response with time.

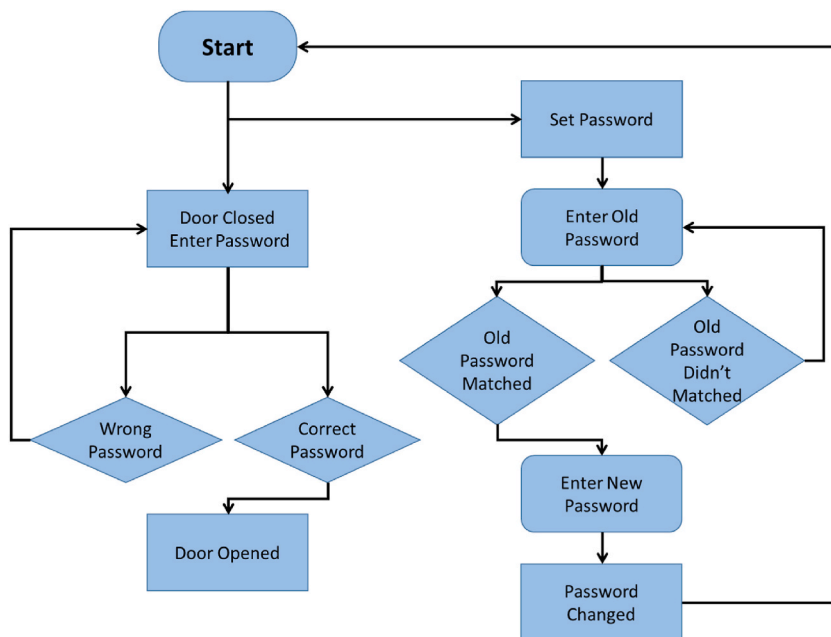


Fig. 6. Flow chart of algorithm operating the DLS with developed sensors system.

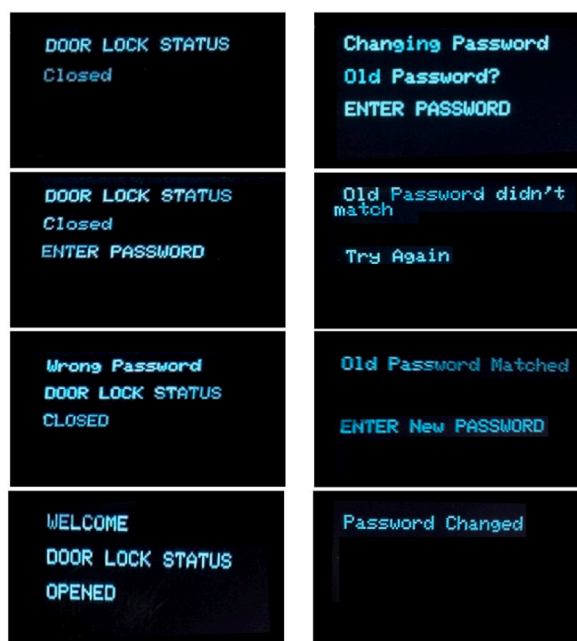


Fig. 7. Pictures of OLED displaying the door lock status and message for the user to guide about the next step.

password has been entered as 1,2,3,4. Supplementary Video S4 is attached to demonstrate the working of the system after protecting it through 80 grams A4 paper. Supplementary Video S5 demonstrate the working of system when wrong key is touched, and system will declare it as wrong password.

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.heliyon.2023.e13586>.

The user can select the password from the 24 different combinations of four numeric sensors. The system is flexible to increase the total number of possible options and it can be increased either by incorporating more touch sensors. The system has shown reliable working for the DLS system. The OLED was connected with the sensor system to show the lock's status for users' guidance. The OLED displayed a message for the users on what to do next, especially in the case of the password reset option. Fig. 7 shows the pictures of different messages displayed on OLED while testing to open the DLS or change the password.

4. Conclusion

The DIY-based invisible and smart DLS is explored in this work. The system consists of a paper-based hybrid geometry sensors keypad for locking/unlocking the door and resetting the password. The sensors are fabricated from cost-effective and disposable materials using the garage-based fabrication process. The system performed equally good for bare and invisible configuration. The invisibility of the sensors lessens the chances of password leak while entering the password. The proposed invisible sensors can find ready applications in cardless and non-keys-based door locking systems for security applications in homes, banks, and lockers.

Declarations

Author contribution statement

Muhammad Qasim Mehmood: Conceived and designed the experiments; Analyzed and interpreted the data; Wrote the paper. Muhammad Shumail Malik: Performed the experiments; Analyzed and interpreted the data; Wrote the paper. Muhammad Hamza Zulfiqar: Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Wrote the paper. Muhammad Atif Khan: Analyzed and interpreted the data; Wrote the paper. Muhammad Zubair, Yehia Massoud: Contributed reagents, materials, analysis tools or data.

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Data availability statement

Data included in article/supp. Material/referenced in article.

Declaration of competing interest

The authors declare no competing interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.heliyon.2023.e13586>.

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